

Internal Assessment: Mathematics, Analysis and Approaches

Inversion in Combinatorial Geometry.

Developing the technique and solving problems from the Baltic Way and the IMO

Shortlist

Word count: -1

1 Introduction

2 Problem Statements

Problem 1 (Baltic Way 2021). We are given 2021 points on a plane, no three of which are collinear. Among any 5 of these points, at least 4 lie on the same circle. Is it necessarily true that at least 2020 of the points lie on the same circle?

Problem 2 (IMO shortlist 1999). A circle is called a *separator* for a set of five points in a plane if it passes through three of these points, it contains a fourth point in its interior, and the fifth point outside the circle.

Prove that every set of five points such that no three are collinear and no four are concyclic has exactly four separators.

Problem 3 (Baltic Way 2019). There are 2019 points given in the plane. A child wants to draw k (closed) discs in such a manner, that for any two distinct points there exists a disc that contains exactly one of these two points. What is the minimal k , such that for any initial configuration of points it is possible to draw k discs with the above property?

Problem 4 (Baltic Way 2024). There is a set of $N \geq 3$ points in the plane, such that no three of them are collinear. Three points A, B, C in the set are said to form a *Baltic triangle* if no other point in the set lies on the circumcircle of triangle ABC . Assume that there exists at least one Baltic triangle.

Show that there exist at least $\frac{N}{3}$ Baltic Triangles.

3 The Concept of Inversion

Out of nothing I have created a strange new universe.

János Bolyai

These problems all involve circles and points, and the simplest solution to each problem uses the common technique of inversion.

3.1 Motivation for Inversion

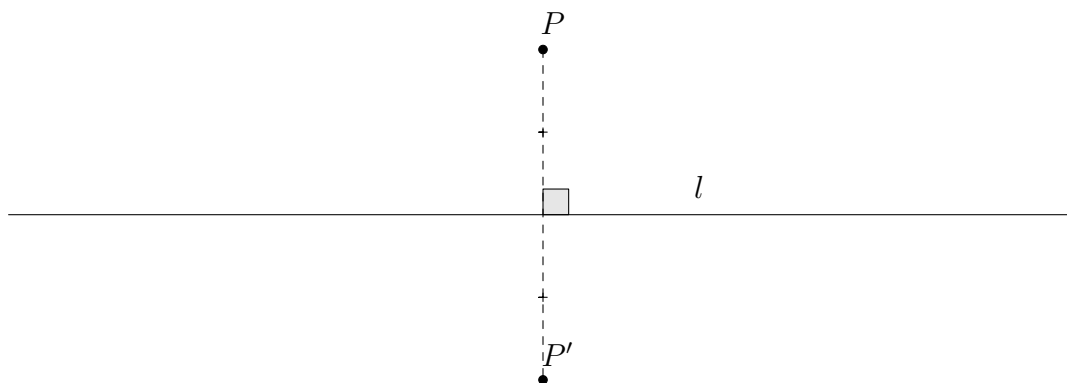


Figure 1: Reflecting a point over a line

Firstly, let us intuitively motivate the upcoming definition of an inversion. We can easily reflect a point across a given line: draw a normal from point onto the line, and mark the reflection as the point on this normal which has an equal distance to the line. In **Figure 1** point P' is the reflection point P across line l .

Now, what if instead of a line we had a circle instead? How should reflection be defined in that case? This premise may seem odd, but it will turn out to be a surprisingly powerful tool if defined appropriately.

Let us first try to define reflection over a circle in ‘the obvious way’: Like with the line, draw a normal from the point onto the circle, and mark the reflection as the point on this normal which has an equal distance to the point where the normal intersects the

circle. This is quite simple to do, since the normal to the circle from a point is simply the line connecting it with the centre.

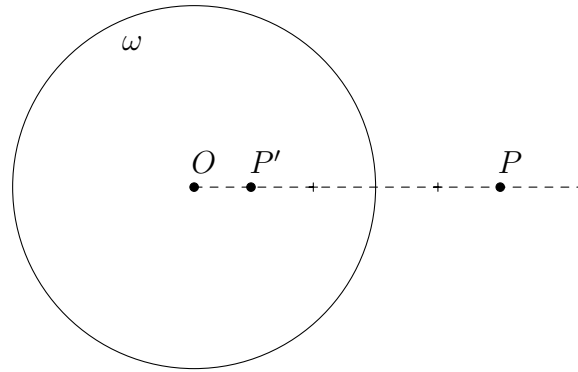


Figure 2: Naïvely reflecting over a circle

In **Figure 2**, point O is the centre of circle ω , and line OP is the normal drawn from point P . With this definition, point P' is the reflection of P across ω . Though this looks reasonable so far, upon attempting to make reflections for more points, we realise the drawbacks of such a guileless definition.

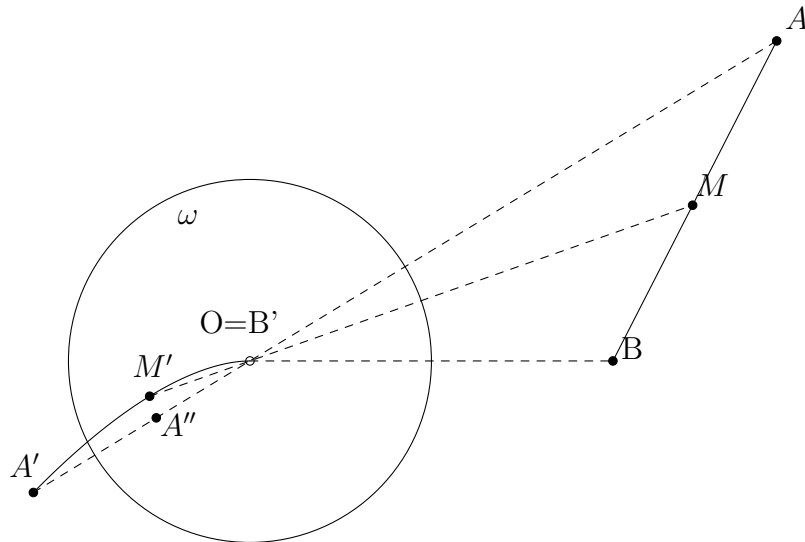


Figure 3: Exposing the weakness of the naïve definition

Again, points A', B' are the reflections of A, B , and likewise A'', B'' are the reflections of A', B' . But making observations from **Figure 3** we realise some undesirable implications of this definition.

It feels wrong that A and A' lie on the same side of the circle (the two sides of a circle are ‘inside’ and ‘outside’) even though A' is the reflection of A . Moreover, though points A' and B' are the reflections of A and B , points A and B themselves are not the reflections of A' and B' . What even is the reflection of O ? It is not possible to define it uniquely since it could be point B , or that point rotated with any angle. Worse still, if point M moves along the segment AB , then its reflection traces some complicated curve with endpoints A', B' , instead of something familiar like a line or a circle.

This last issue in particular cripples this definition of reflecting over a circle. When we reflect many points over a line, they still stay the same shape: lines map to lines, circles map to circles; it is not possible for simple things to have complicated reflections.

We need to define the reflection over a circle in a more deliberate way to avoid these undesirable weaknesses. Since we call the operation ‘reflection’, it should share some of the most fundamental properties fulfilled when reflecting over a line.

Property 1. The reflection of each point exists and is uniquely defined.

Property 2. Reflecting the reflection of a point yields the point itself.

Property 3. Points on the line l are fixed, i.e. reflect to themselves.

Property 4. Points not on the line l are reflected to the opposite side of l .

We can take these properties directly as the requirements for our definition of reflecting over a circle, just by replacing line l with circle ω . We have commented on how the naïve definition fails to satisfy **Properties 1, 2, and 4**, but **Property 3** is satisfied.

Our note about the reflection of segment AB in **Figure 3**, however, warrants another property to be demanded from the upcoming definition:

Property 5. If a point moves along a circle or a line, then its reflection also traces a circle or a line.

We shall introduce *inversion*, which is a possible definition of reflection across a circle satisfying all these properties. In fact, the words are synonymous by convention, and the author apologises for his abuse of terminology in the previous section.

3.2 Definition of Inversion

The author finds no better way to formulate of the definition of inversion than simply taking an extended quotation from his favorite mathematics textbook, *Euclidean Geometry in Mathematical Olympiads* (2016).

The idea is to view every line as a circle with infinite radius. We add a special point P_∞ to the plane, which every ordinary line passes through (and no circle passes through). This is called the **point at infinity**. Therefore, every choice of three distinct points determine a unique circle or line—three ordinary points determine a circle, while two ordinary points plus the point at infinity determine a line.

With this said, we can now define an inversion. Let ω be a circle with centre O and radius R . We say an **inversion** about ω is a map (that is, a transformation) which does the following.

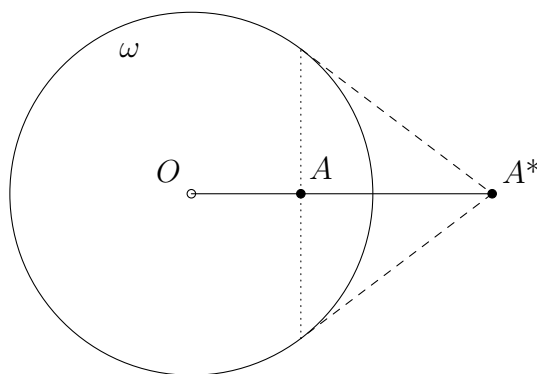


Figure 4: A^* is the image of point A when we take an inversion about ω

- The centre O of the circle is sent to P_∞
- The point P_∞ is sent to O .
- For any other point A , we send A to the point A^* lying on ray OA such that $OA \cdot OA^* = r^2$.

Quoted at length from [1]

The definition of P_∞ above facilitates taking an inverse of point O and allows us to treat lines and circles as highly similar objects.

Figure 4 shows a geometrical interpretation of the relationship between a point A and its inverse A^* : A is the point where the chord given by the tangents of A^* to ω intersects line OA^* . A proof will be omitted as it is not central to our discussion; the reader is encouraged to prove the statement.

3.3 Properties of Inversion

Let us verify that inversion satisfies all the properties demanded in the previous section. Consider a point on the plane A , and its inverse A with respect to a circle of centre O and radius r .

Property 1.: If $A = O$, we explicitly defined its inverse as P_∞ . Otherwise we can construct A^* , by drawing ray OA and marking A^* as the point on there whose distance to O is $\frac{r^2}{OA}$.

Property 2.: If $A = O$, our explicit definitions guarantee the inverse of its inverse is also O . Otherwise we have

$$\begin{cases} OA \cdot OA^* = r^2 \\ OA^* \cdot O(A^*)^* = r^2 \end{cases} \quad (3.1)$$

so $OA = O(A^*)^*$. Since A^* and $(A^*)^*$ are both on ray OA , we necessarily have $A = (A^*)^*$.

Property 3.: We only consider points A which lie on the circle. This simply means $OA = r$, so trivially $OA^* = r$ and $A = A^*$.

Property 4.: If point A is inside the circle, then $OA < r$, so $OA^* = \frac{r^2}{OA} > \frac{r^2}{r} = r$, so its inverse is outside. Likewise, if point A is outside, we have $OA > r \implies OA^* < r$, so its inverse is inside, as desired.

Property 5.: We show that a line or circle inverts to a line or a circle, by proving a more specific claim about their behaviour upon inversion.

Property 5.6. A line passing through O inverts to itself.

Property 5.7. The inverse of a circle not passing through O is a circle not passing through O .

Property 5.8. The inverse of a line not passing through O is a circle passing through O . Equivalently, the inverse of a circle passing through O is a line not passing through O .

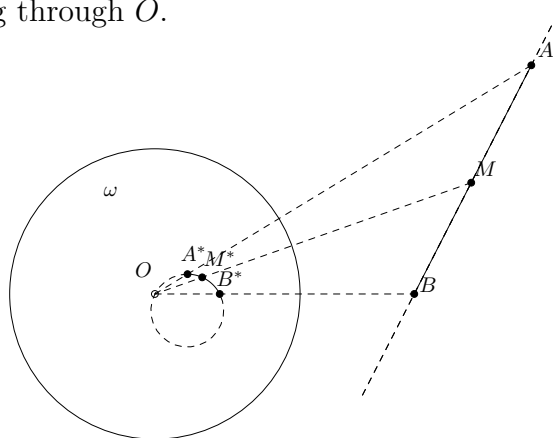


Figure 5: Same as **Figure 3**, but using inversion.

We see **Property 5.8.** in action in **Figure 5**. The inverse of line AB is the circle passing through O , A^* , and B^* .

Verifying **Property 5.6.** is trivial. The inverse of each point on the line is still on the line, since the line goes through O .

For **Property 5.7.**, consider the following proof. (I can either give a geometrical proof, or show the transformation on the complex plane which achieves inversion, and then prove this algebraically. Alternatively I could just omit a proof.) Define the function $I : \mathbb{C} \setminus \{0\} \rightarrow \mathbb{C} \setminus \{0\}$ as $I(z) = \overline{z^{-1}}$

Property 5.8. has a similar proof.

These three properties combined are of crucial importance in the solution of our problems in **Section 2**. To supplement them, let us formulate and derive an additional property not mentioned in Chen's masterpiece (perhaps he takes it to be trivial?)

Lemma 1. *If points A, B are on different sides of circle or line Ξ , then their inverses will be on different sides of the inverse of Ξ .*

We can define the notion of two points A, B lying on different sides of an object Ξ formally, to mean that all (continuous) paths starting at A and ending at B on the plane intersect Ξ at least once. By this definition, A and A' of **Figure 3** are clearly on the same side of ω , since we can draw a curve connecting them which never touches ω .

Proof. Assume, for contradiction, that A^* and B^* are on the same side of the inverse of Ξ . This means there exists a path from A^* to B^* which never intersects the inverse of Ξ . Then the inverse of this path must never intersect Ξ , by **Property 1**, which is contradiction: it would imply A and B are on the same side!

4 Solutions

Actually, in none of these problems matters the radius of the circle through which inversion is performed.

4.1 Solving Problem 1.

Problem 5 (Baltic Way 2021). We are given 2021 points on a plane, no three of which are collinear. Among any 5 of these points, at least 4 lie on the same circle. Is it necessarily true that at least 2020 of the points lie on the same circle?

Answer: Yes.

Let us label the points $A_1, A_2, \dots, A_{2021}$. If all points lie on a circle, there is no need

to discuss any further. Otherwise, suppose without loss of generality, that among points A_1, A_2, A_3, A_4, A_5 , the latter four points A_2, A_3, A_4, A_5 lie on the same circle γ , but A_1 is not on γ .

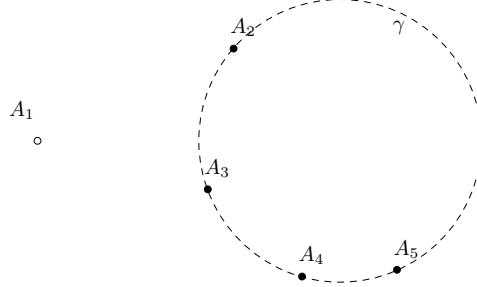


Figure 6: $A_2A_3A_4A_5$ lie on a circle γ not containing A_1 .

Next, let A_i^* denote the inverse of point A_i , for $i = 2, 3, \dots, 2021$, with respect to the circle centered at A_1 with radius 1. What does the problem statement say about these inverses?

We are guaranteed that when we select any five points from $A_1, \dots, A_{2021}$, at least four of them lie on the same circle. In the case that these five points are A_1 , and four others, call them A_i, A_j, A_k, A_l , we can actually say that

1. All four of $A_i^*, A_j^*, A_k^*, A_l^*$ lie on a circle not containing A_1 , by [Property 5.7.](#), or
2. At least three of $A_i^*, A_j^*, A_k^*, A_l^*$ lie on a line not containing A_1 , by [Property 5.8.](#)

Here, the first case represents the situation when A_1 is not one of the four points which lie on the same circle, while the second case represents the situation when it is.

By [Property 5.7.](#), $A_2^*A_3^*A_4^*A_5^*$ lie on a circle, which we shall call γ^* . We wish to show that all 2020 inverses, including these four, lie on γ^* .

Suppose, for contradiction, and without loss of generality, that point A_6^* is not on the γ^* . Then, by observing points $A_2^*A_3^*A_4^*A_6^*$, we get that A_6^* must lie on one of the lines $A_2^*A_3^*$, $A_2^*A_4^*$, $A_3^*A_4^*$. Suppose, without loss of generality, that A_6^* lies on line $A_2^*A_3^*$. Next,

observe the points $A_3^*A_4^*A_5^*A_6^*$. Again A_6^* must lie on one of the lines $A_3^*A_4^*$, $A_3^*A_5^*$, $A_4^*A_5^*$. Yet it can't be on $A_3^*A_4^*$ or $A_3^*A_5^*$, since that would imply $A_6^* = A_3^*$. Hence A_6^* lies on $A_4^*A_5^*$. This means A_6^* is exactly the intersection of lines $A_2^*A_3^*$ and $A_4^*A_5^*$.

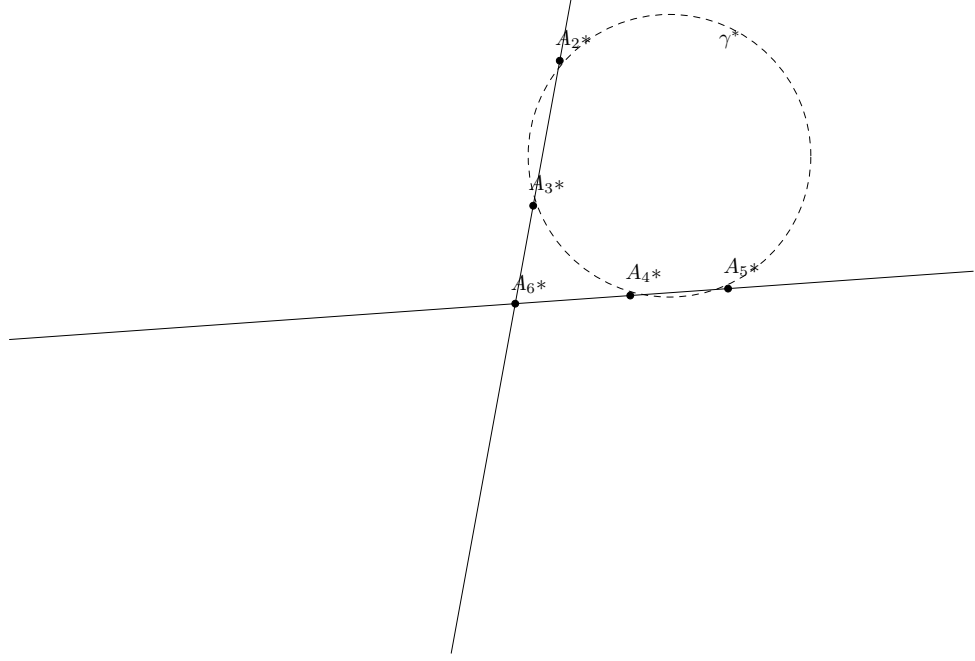


Figure 7: A_6^* is the intersection of lines $A_2^*A_3^*$ and $A_4^*A_5^*$.

Consider the case that among the 2020 inverses, there existed a point A_k^* differing from $A_2^*A_3^*A_4^*A_5^*A_6^*$ that lies on the circle γ^* . Then we could observe $A_3^*A_4^*A_6^*A_k^*$ for contradiction: they cannot all lie on the same circle, and clearly no three of them may be collinear or otherwise two distinct points would be equal.

Otherwise none of A_7^*, A_8^*, A_9^* lie on γ^* . But then similarly to how we showed that A_6^* is the intersection of $A_2^*A_3^*$ and $A_4^*A_5^*$, we get that A_7^*, A_8^*, A_9^* are all intersections of two lines formed by points $A_2^*A_3^*A_4^*A_5^*$.

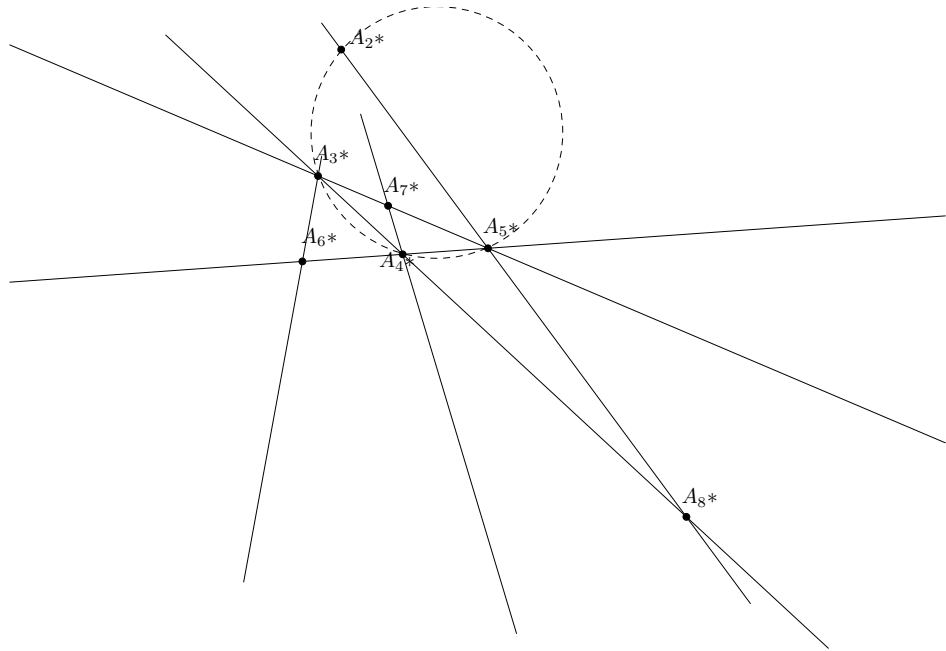


Figure 8: A_6^*, A_7^*, A_8^* occupy the valid intersections, and A_9 causes contradiction.

That's impossible, since there are only three such intersections. Hence we have contradiction, which means point A_6^* , and all other inverses too, must lie on γ^* , and we are done.

4.2 Solving Problem 2.

4.3 Solving Problem 3.

4.4 Solving Problem 4.

5 Conclusions

References

- [1] Evan Chen. *Euclidean Geometry in Mathematical Olympiads*. MAA Problem Books, 311 pp. Washington, DC: Mathematical Association of America, 2016. ISBN: 9780883858394.